

GeCCo Presentation Script — Slides 1–7

Keyword Version · ~10 minutes total · Rmus et al. (2025)

Slide 1 – Opening

≈1 min

- Welcome, name paper: Rmus et al. 2025, GeCCo, NeurIPS
- Presenters: Yingzhu Chen & Christian Block
- Frame: not a full summary — an argument-driven reading
- Core message: LLMs can speed up cognitive model discovery, but better fit $\neq$ better explanation
- Transition: why is cognitive modeling hard in the first place?

Slide 2 – Problem

≈1.5 min

- Traditional modeling loop: theory $\rightarrow$ code $\rightarrow$ fit $\rightarrow$ revise
- Slow, expert-dependent
- Requires: literature review, formal theory, implementation, fitting — repeat
- Real risk: narrow model space — only test what we already know how to write
- Transition: so what exactly is a "cognitive model"?

Slide 3 – Concept

≈1.5 min

- Definition: psychological idea $\rightarrow$ equations/code $\rightarrow$ predictions
- Example: "people learn from rewards" $\rightarrow$ Python function $\rightarrow$ choice probabilities
- Show template briefly: `learning_rate`, `beta`, `log-likelihood`
- Key point: different mechanisms can produce same behavior $\rightarrow$ need model comparison
- Transition: GeCCo automates part of this model-building

Slide 4 – Method: the loop

≈1.5 min

- Core loop: Prompt $\rightarrow$ LLM writes 3 model codes $\rightarrow$ fit on held-out data $\rightarrow$ select by BIC $\rightarrow$ feedback
- LLM proposes, doesn't replace evaluation
- Data + BIC select among candidates
- Key phrase: "guided code generation, not one-shot prompting"
- Transition: prompt itself is part of the experimental design

Slide 5 – The prompt as method

≈1.5 min

- 5 ingredients: task description, behavioral data, guardrails, code template, feedback
- Point: prompting here isn't casual chatting — it constrains the search space
- Benefit: outputs are executable & comparable
- Cost: less creativity, risk of template bias
- Transition: once models are generated, how is "better" decided?

Slide 6 – Evaluation criteria

≈1.5 min

- Three tools: NLL (surprise at real choices), BIC (fit + parameter penalty), PPC (can simulated data look human?)
- Intuition only, no formulas — formulas in backup
- Critical point: lower BIC = better model comparison, NOT proof of psychological truth
- Transition: does this approach generalize?

Slide 7 – Evidence across domains

≈1.5 min

- Four domains tested: decision making (pWADD), learning (RW4 α), planning (Hybrid), working memory (RL-WM)
- Result headline: LLM models matched or outperformed all four baselines
- Note: planning result favored by EXP, not significant BIC difference — be precise here
- Transition: will deep-dive decision making and learning as representative cases, then summarize planning/WM

Total runtime target: ~10 minutes for slides 1–7.